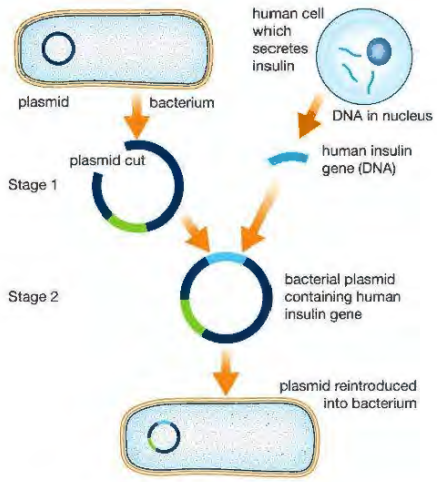
SKILLS ANALYSIS 8

5 The diagram shows the main stages in transferring the human insulin gene to a bacterium.



a Name the enzymes used at stages 1 and 2.

SKILLS CRITICAL THINKING 8

b What is the role of the plasmid in this procedure?

c How would the insulin-producing bacteria be used to produce significant amounts of insulin?

SKILLS REASONING 9

d Why is the insulin produced this way preferred to insulin extracted from other animals?

SKILLS INTRAPERSONAL 12

6 Carry out some research to find out more about the use of transgenic animals for organ transplantation. What are the arguments for and against its use? What are the ethical objections?

SKILLS CRITICAL THINKING 8

7 Producing genetically modified plants and animals is more complex than producing genetically modified bacteria.

a Describe two ways in which genes can be introduced into plant cells.

b How are these genetically modified cells used to produce whole organisms?

c What sort of animal cell is genetically modified and then used to produce the whole organism?

SKILLS INTRAPERSONAL 12

8 Write an essay about the importance of genetic engineering. In your essay you should make reference to:

a important potential benefits resulting from genetic engineering in animals and plants

b concerns about the risks resulting from genetic engineering in animals and plants.

UNIT QUESTIONS

SKILLS CRITICAL THINKING 8 **1**

Some experimental sheep were genetically modified, using a human gene. The gene codes for a protein involved in blood clotting, called factor IX.

The gene for factor IX was cut from human DNA and injected into a sheep body cell. The nucleus of this body cell was transferred to an enucleated sheep ovum. The ovum was cultured in a dish and then implanted in the uterus of a surrogate sheep to complete its development.

Two lambs produced by this method grew to maturity. They produced milk from which Factor IX could be extracted.

a Name an enzyme that can be used to cut DNA. **(1)**

7 8

b What is an *enucleated ovum*? **(2)**

c The sheep produced by this method are called *transgenic*. What is meant by this term? **(1)**

SKILLS REASONING 7

d The transgenic sheep were reproduced by *cloning*. What are the advantages of producing the sheep in this way? **(3)**

6

e Factor IX can be used to treat people with a blood clotting disorder called haemophilia. Why is important to have blood that clots? **(2)**

SKILLS CRITICAL THINKING 5

f Blood contains plasma, white blood cells, red blood cells and platelets. Which *two* components of the blood are involved with blood clotting? **(2)**

Total 11 marks

SKILLS CRITICAL THINKING 9 **2**

Some genes are transferred from one plant species to another. These genes are called 'jumping genes'. Environmentalists are concerned that genetically modified plants may transfer some of their genes to wild species. Explain their concern about genetically modified plants that:

a have genes that make them resistant to herbicides (weedkillers) **(2)**

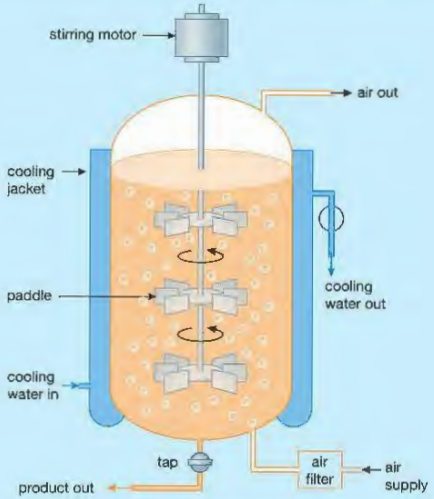
b have genes that make them resistant to pests **(2)**

c have genes that increase the yield of the crop they can produce. **(2)**

Total 6 marks

SKILLS REASONING

3 The diagram shows a fermenter of the type used to grow microorganisms used to produce human insulin. Insulin is used to treat people with diabetes.



a Explain why air is pumped through the fermenter. **(2)**

b Why is it necessary to keep the temperature in the fermenter constant? **(2)**

c Suggest one other condition that must be kept constant inside the fermenter. **(1)**

d The fermenter is sterilised by steam. Suggest why this is better than using disinfectants. **(2)**

e Insulin can also be obtained from animal pancreases. Suggest two advantages of using microorganisms to produce the insulin needed by diabetics. **(2)**

Total 9 marks

SKILLS CRITICAL THINKING

4 a Explain the meaning of the following terms:
i a plasmid
ii a bacteriophage. **(6)**

b Explain the importance of the following enzymes in genetic engineering:
i restriction enzymes
ii ligase enzymes. **(6)**

Total 12 marks

5 Two organisms used in beer production are Saccharomyces cerevisiae and barley.

a What type of organism is Saccharomyces cerevisiae? **(1)**

SKILLS INTERPRETATION

b 'The respiration of Saccharomyces cerevisiae is essential to the production of beer.' Explain this statement. Include a chemical word equation in your answer. **(3)**

SKILLS CRITICAL THINKING

c Explain the role of barley in beer production. **(3)**

Total 7 marks

APPENDIX A: A GUIDE TO EXAM QUESTIONS ON EXPERIMENTAL SKILLS

Copies of official specifications for all Edexcel qualifications may be found on the Edexcel website, www.edexcel.com. Past papers, marks schemes and examiners' reports are also available on the website.

WHY IS THIS APPENDIX IMPORTANT?

This appendix is designed to help you gain the 20% of marks allotted to the questions on experimental skills. These skills are tested in both Paper 1 and Paper 2, mixed up with theory questions. The practicals that you should know about are included in the biology content of the specification and fully described in this book. In the exams you will be asked questions based on these, although the apparatus shown may be slightly different. However, you will also be given questions about unfamiliar biological investigations. Don't be worried by these – you are not expected to have carried them out, just to be able to apply your knowledge and understanding to interpret experimental design or to analyse data.

Questions in the exams will generally cover the following areas:

- understanding safety precautions
- recognising apparatus and understanding how to use it
- recalling tests for certain substances
- manipulating data
- plotting graphs
- understanding experimental design
- planning experiments.

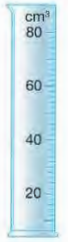
1. SAFETY PRECAUTIONS

As part of a question, you may be asked to comment on appropriate safety precautions. The most obvious precaution is to **wear eye protection**. Eye protection must be worn whenever chemicals or Bunsen burners are used. This will apply to many experiments or investigations, with a few exceptions such as fieldwork. Some other examples of safety precautions, and the reason they are taken, are:

Precaution	Reason
wash hands after handling biological material, such as plants, soil etc.	to avoid contamination
keep flammable liquids such as ethanol away from a naked flame	to avoid the ethanol catching fire
take care with fragile glassware such as pipettes, microscope cover slips, etc.	to avoid cutting yourself
do not touch electrical apparatus (e.g. a microscope with built-in lamp) with wet hands	to avoid getting an electric shock
use a water bath to heat a test tube of water, rather than heating it directly in a Bunsen flame	to avoid the heated liquid jumping out of the tube

2. RECOGNISING APPARATUS AND DEMONSTRATING AN UNDERSTANDING OF HOW TO USE THEM

One of the simplest types of question in the exams will require you to recognise common pieces of laboratory apparatus, such as a Bunsen burner, thermometer, measuring cylinder and stopwatch. You will also need to recognise particular 'biological' apparatus such as quadrat, microscope, bench lamp and so on, and how they are used. If you have been able to carry out most of the practical work in the specification, this kind of question is very straightforward, for example:



(a) What is the name of this piece of apparatus? **(1 mark)**

(b) Draw a line on the apparatus to show a volume of 30 cm^3 **(1 mark)**

Note: Use a pencil to draw any lines on diagrams, graphs etc. Then you can rub it out if you make a mistake. On this measuring cylinder a horizontal line should be drawn half way between the 20 and 40 cm^3 marks – it's that easy!

3. RECALLING TESTS FOR SUBSTANCES

There are really only five chemical tests in the specification. These are for:

1. starch – using iodine solution
2. glucose (or 'reducing sugar') – heating with Benedict's solution
3. protein – using the biuret test
4. fat (lipid) – using the emulsion test
5. carbon dioxide – using hydrogen carbonate indicator.

You should know the starch test (page 58) and how leaves can be tested for starch (page 136).

The tests for glucose, protein and fat are on page 58 The effects of carbon dioxide on hydrogen carbonate indicator are described on page 141.

Once you have learnt these, the questions are easy! For example:

John decided to test some milk for glucose.